

Testing with one real parameter

Outline

- 1) One-sided tests in general
- 2) Two-sided tests
- 3) UMP unbiased tests

One-sided tests in general

$$\mathcal{P} = \{P_\theta: \theta \in \Theta \subseteq \mathbb{R}\}, \quad \theta_0 \in \Theta$$

$H_0: \theta \overset{\geq}{\leq} \theta_0$ vs $H_1: \theta \overset{<}{>} \theta_0$ called one-sided hypothesis

Often, no UMP test exists.

LRT may vary for different θ_1 values

If n large, could prioritize $\theta_1 = \theta_0 + \varepsilon$, $\varepsilon \downarrow 0$

$$\log LR(x) = \log \frac{p_{\theta_0 + \varepsilon}(x)}{p_{\theta_0}(x)} \approx \varepsilon \cdot \dot{\ell}(\theta_0; x)$$

$\Rightarrow$ Use score at θ_0 $\dot{\ell}(\theta_0; x)$ as test stat.

$$\phi(x) = 1\{\dot{\ell}(\theta_0; x) \geq c_\alpha\}$$

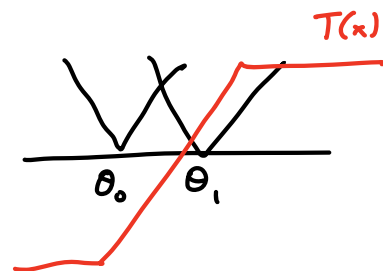
Need to check $\beta_\phi(\theta) \leq \alpha$ for $\theta \leq \theta_0$

Ex. Laplace: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{2} e^{-|x-\theta|}$

Test $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$

$$\begin{aligned} \theta_1 > \theta_0: \log(p_{\theta_1}(x)/p_{\theta_0}(x)) &= \sum_{i=1}^n |x_i - \theta_0| - |x_i - \theta_1| \\ &= \sum T(x_i) \end{aligned}$$

$$T(x) = \begin{cases} \theta_0 - \theta_1 & x \leq \theta_0 \\ 2x - \theta_0 - \theta_1 & \theta_0 \leq x \leq \theta_1 \\ \theta_1 - \theta_0 & x \geq \theta_1 \end{cases}$$



Score $\dot{\ell}(\theta_0; X) = \frac{d}{d\theta} \sum_i -|x_i - \theta| \big|_{\theta=\theta_0}$

$$= \sum_{i=1}^n \text{sign}(x_i - \theta_0)$$

Equivalent: $S(X) = \sum_{i=1}^n \mathbb{1}\{x_i \geq \theta_0\}$ Sign test

$$\begin{aligned} &\sim \text{Binom}(n, P_{\theta}(x_i \geq \theta_0)) \\ &\stackrel{\theta=\theta_0}{=} \text{Binom}(n, 1/2) \end{aligned}$$

Nonparametric example $X_i \stackrel{iid}{\sim} F, \quad \theta(F) = \text{median}(F)$

Test $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$

$$S(X) \sim \text{Binom}(n, 1 - F(\theta_0)) = \text{Binom}(n, 1/2) \quad \text{if } \theta(F) = \theta_0$$

Stochastically incr.

Def A real-valued statistic $T(X)$ is stochastically increasing in Θ if

$P_{\Theta}(T(X) \leq t)$ is non-incr. in Θ , $\forall t$

If $\phi(x)$ rejects for large $T(X)$:

$$\phi(x) = 1\{T(X) > c\} + \gamma 1\{T(X) = c\}$$

and $T(X)$ is stochastically increasing in Θ ,

$$E_{\Theta} \phi(X) = (1-\gamma) P_{\Theta}(T > c) + \gamma P_{\Theta}(T \geq c) \nearrow \text{in } \Theta$$

E_X $X_i \stackrel{\text{iid}}{\sim} \rho(x-\theta)$ (location family)
 $T(X)$ = sample mean, median, sign statistic

E_X $X_i \stackrel{\text{iid}}{\sim} \frac{1}{\theta} \rho(x/\theta)$ (scale family)
 $T(X)$ = $\sum X_i^2$ or median($|X_1|, \dots, |X_n|$)

Two-sided Alternatives

Setup: $\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, $\theta_0 \in \Theta^0$

Test $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$

(Can be generalized naturally to $H_0: \theta \in [\theta_1, \theta_2]$)

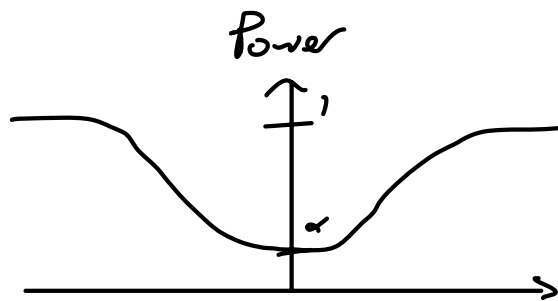
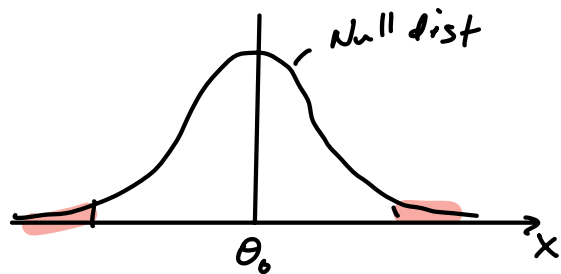
Two-tailed test rejects when $T(X)$ is "extreme"

$$\phi(x) = \begin{cases} 1 & T(X) > c_2 \text{ or } T(X) < c_1 \\ 0 & T(X) \in (c_1, c_2) \\ \gamma_i & T(X) = c_i \end{cases}$$

Two ways to reject. How to balance?

For symmetric distributions like $N(\theta, 1)$,
natural choice is to equalize "lobes" of rej. region

$$\phi_2(x) = 1\{|x - \theta_0| > z_{\alpha/2}\} \text{ for } H_0: \theta = \theta_0$$



For asymmetric dists, or interval null $H_0: \theta \in [\theta_1, \theta_2]$,
more complicated

Equal-tailed & unbiased tests

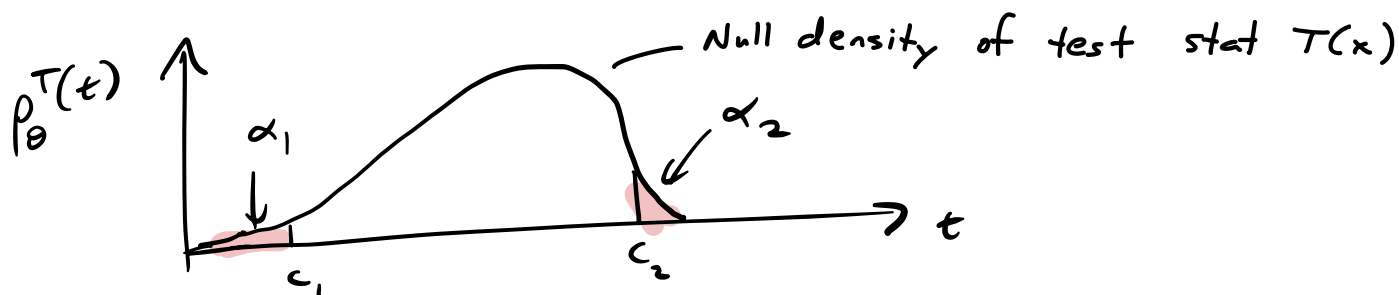
Point null ($H_0: \theta = \theta_0$)

$$\text{Let } \alpha_1 = P_{\theta_0}(T < c_1) + \gamma_1 P_{\theta_0}(T = c_1)$$

$$\alpha_2 = P_{\theta_0}(T > c_2) + \gamma_2 P_{\theta_0}(T = c_2)$$

Valid if $\alpha_1 + \alpha_2 = \alpha$ (α_1 is "free parameter")

Idea 1: Equal-tailed test : $\alpha_1 = \alpha_2 = \frac{\alpha}{2}$



Ex $X \sim \text{Exp}(\theta)$, test $H_0: \theta = 1$

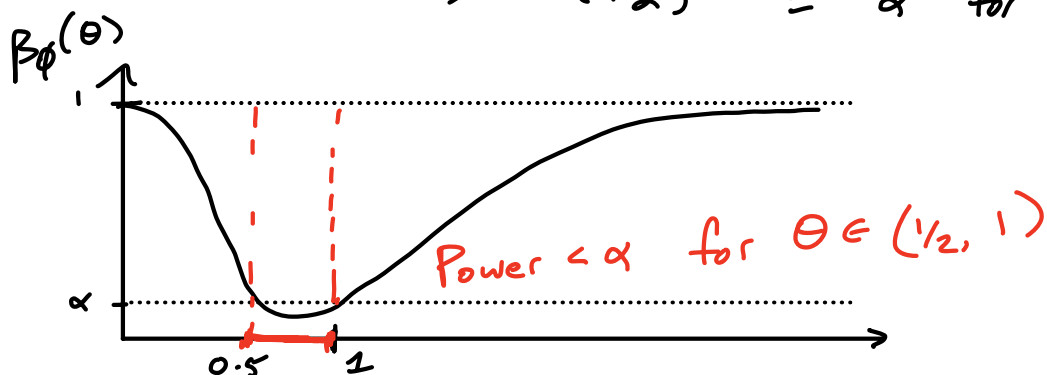
$$\text{Solve for cutoffs: } \frac{\alpha}{2} = P_1(X \leq c_1) = 1 - e^{-c_1} \Rightarrow c_1 = -\log(1 - \frac{\alpha}{2})$$

$$1 - \frac{\alpha}{2} = 1 - e^{-c_2} \Rightarrow c_2 = -\log(\frac{\alpha}{2})$$

$$\phi(x) = 1\{X < -\log(1 - \frac{\alpha}{2})\} + 1\{X > -\log(\frac{\alpha}{2})\}$$

$$\beta_{\phi}(\theta) = P_{\theta}\left\{\frac{X}{\theta} < \frac{-\log(1 - \frac{\alpha}{2})}{\theta}\right\} + P_{\theta}\left\{\frac{X}{\theta} > \frac{-\log(\frac{\alpha}{2})}{\theta}\right\}$$

$$= 1 - (1 - \frac{\alpha}{2})^{1/\theta} + (\frac{\alpha}{2})^{1/\theta} = \alpha \text{ for } \theta = 1 \text{ or } 1/2$$



Unbiased tests

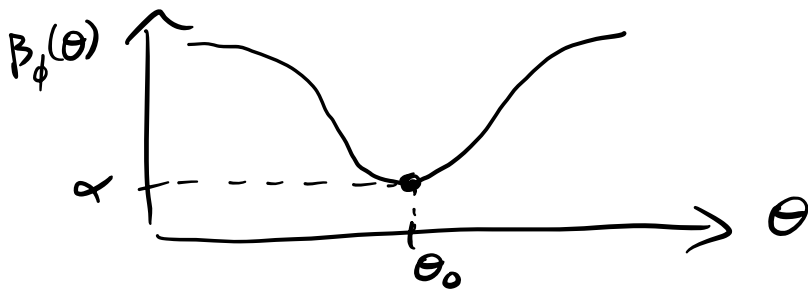
Def $\phi(x)$ is unbiased if $\inf_{\theta \in \Theta} \mathbb{E}_{\theta} \phi(x) \geq \alpha$

Idea 2: Unbiased test: ensure $\min_{\theta} \beta_{\phi}(\theta) = \alpha$

Choose c_1, γ_1 and c_2, γ_2 to solve:

$$\beta_{\phi}(\theta_0) = \alpha \quad (2 \text{ equations, "2" unknowns})$$

$$\frac{d\beta_{\phi}}{d\theta}(\theta_0) = 0$$



Ex: 1-parameter exp. family, $H_0: \eta = \eta_0$ vs $H_1: \eta \neq \eta_0$

$$X \sim e^{\eta T(x) - A(\eta)} h(x) \quad (\text{MLR in } T(X))$$

Assume $T(X)$ continuous, solve

$$\alpha = \beta_{\phi}(\eta_0) = \mathbb{P}_{\eta_0}(T < c_1) + \mathbb{P}_{\eta_0}(T > c_2)$$

$$0 = \frac{d\beta_{\phi}}{d\eta}(\eta_0) = \text{Cov}_{\eta_0}(\phi(T), T)$$

$$= \mathbb{E}_{\eta_0}[(\phi(T) - \alpha) T(X)]$$

Theorem Assume $X \sim e^{\theta T(x) - A(\theta)} h(x)$

$$H_0: |\theta - \theta_0| \leq \delta \quad \text{vs} \quad H_1: |\theta - \theta_0| > \delta, \quad \delta \geq 0$$

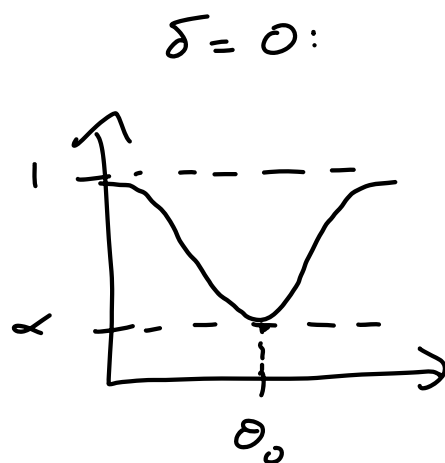
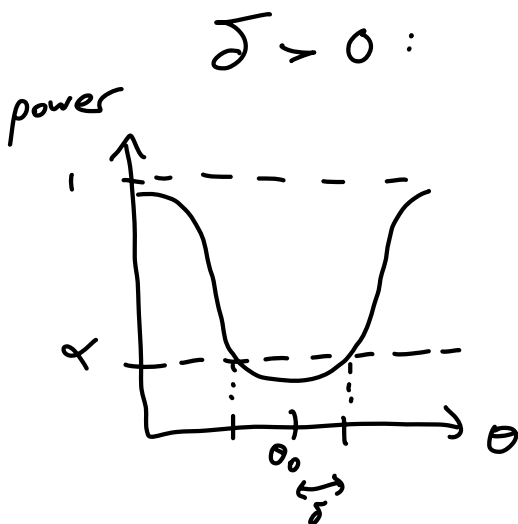
Let ϕ^* be test that rejects for extreme $T(x)$, with $c_1, c_2, \gamma_1, \gamma_2$ chosen so:

$$(i) \beta_{\phi^*}(\theta_0 + \delta) = \beta_{\phi^*}(\theta_0 - \delta) = \alpha$$

and, if $\delta = 0$ (point null)

$$(ii) 0 = \dot{\beta}_{\phi^*}(\theta_0) = \mathbb{E}_{\theta_0}[(T - \mathbb{E}_{\theta_0} T) \phi(x)]$$

Then ϕ^* is UMPU



Proof: Assume wlog $\theta_0 = 0$

($\delta = 0$):

Want to solve

$$\text{maximize } \int \phi p_{\theta} d\mu$$

$$\text{s.t. } \int \phi p_0 d\mu = \alpha \quad \leftarrow (\text{unbiased})$$

$$\int \phi (T - \mathbb{E}_0 T) p_0 d\mu = 0$$

Lagrange form:

$$\max \int \phi (p_{\theta_1} - \lambda_1 p_0 - \lambda_2 p_0 (T - \mathbb{E}_0 T)) d\mu$$

$$= \int \phi \left(\frac{p_{\theta_1}}{p_0} - \lambda_1 - \lambda_2 (T - \mathbb{E}_0 T) \right) dP_0$$

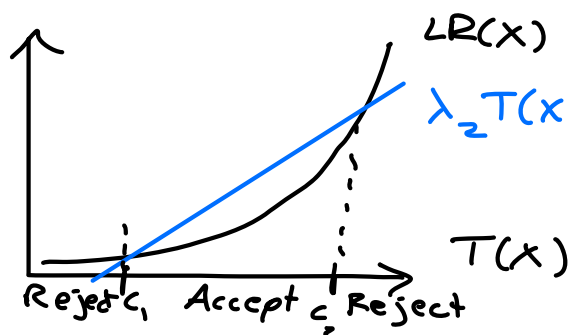
$$\Rightarrow \phi^*(x) = \begin{cases} 1 & LR(x) > \lambda_2 T(x) + \lambda_0 \\ 0 & LR(x) < \lambda_2 T(x) + \lambda_0 \\ \text{arb.} & LR(x) = \lambda_2 T(x) + \lambda_0 \end{cases}$$

$\lambda_1 + \lambda_2 \mathbb{E}_0 T$

$$LR(x) = e^{\theta_1 T(x) + A(0) - A(\theta_1)}$$

$\theta_1 > 0$:

can find λ_1, λ_2
to match c_1, c_2



$\theta_1 < 0$
similar

Suppose ϕ, ϕ^* satisfy constraints,
 ϕ^* maximizes Lagrangian for λ_1, λ_2

$$\begin{aligned} \beta_{\phi}(\theta_1) &= \beta_{\phi}(\theta_1) + \lambda_1 (\beta_{\phi}(0) - \alpha) \\ &\quad + \lambda_2 \dot{\beta}_{\phi}(0) \\ &\leq \beta_{\phi^*}(\theta_1) + \lambda_1 (\beta_{\phi^*}(0) - \alpha) \\ &\quad + \lambda_2 \dot{\beta}_{\phi^*}(0) \\ &= \beta_{\phi^*}(\theta_1) \end{aligned}$$

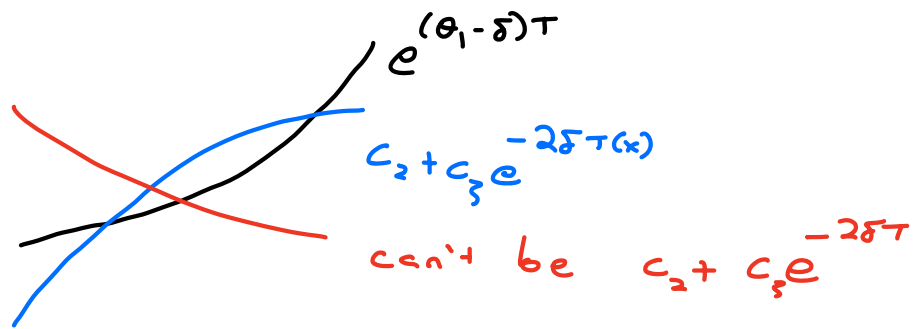
$$\begin{aligned} (\delta > 0) \quad \max \quad & \int \phi \rho_{\theta_1} d\mu \\ \text{s.t.} \quad & \int \phi \rho_{\delta} d\mu = \int \phi \rho_{-\delta} d\mu = \alpha \end{aligned}$$

Lagrangian:

$$\begin{aligned} & \int \phi (\rho_{\theta_1} - \lambda_1 \rho_{\delta} - \lambda_2 \rho_{-\delta}) d\mu \\ &= \int \phi (c_1 e^{\theta_1 T} - c_2 e^{\delta T} - c_3 e^{-\delta T}) dP_0 \quad c_i > 0 \end{aligned}$$

$\theta_1 > \delta$:

Reject for $c_1 e^{(\theta_1 - \delta)T(x)} > c_2 + c_3 e^{-2\delta T(x)}$



Rest of proof same as $\delta = 0$